Ejercicios_Rafa

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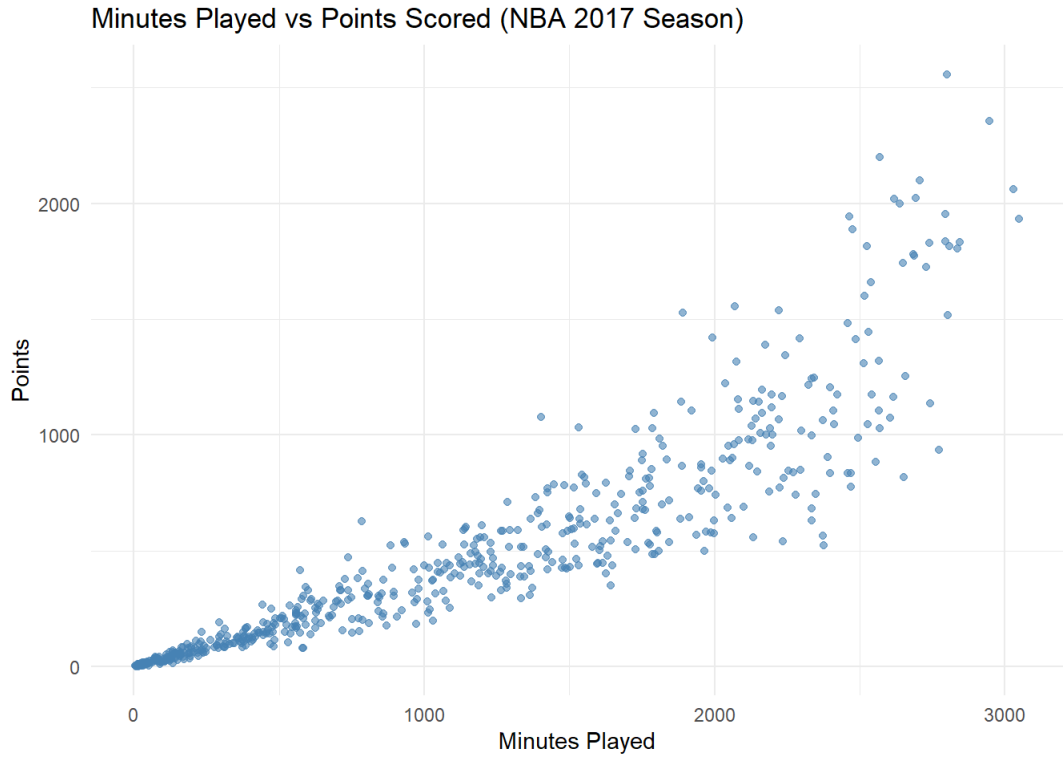
```
library(dplyr)
library(ggplot2)
library(ggpubr)
library(viridis)
library(tidyverse)
library(broom)
library(lmtest)
library(car)
```

Ejercicio 1 - NBA Stats

Minutes played VS points

```
# load data
season <- read.csv("C:/Users/enriq/Desktop/Master/Data Science (MIB)/Clases Rafa/data/NBA/Seasons_Stats.c
sv", header = T, sep = ",")
nba_2017 <- season %>%
  dplyr::filter(Year == 2017) %>%
  dplyr::select(Player, MP, PTS) %>%
  dplyr::filter(!is.na(MP), !is.na(PTS), MP > 0, PTS > 0)


ggplot(nba_2017, aes(x = MP, y = PTS)) +
  geom_point(alpha = 0.6, color = "steelblue") +
  labs(
    title = "Minutes Played vs Points Scored (NBA 2017 Season)",
    x = "Minutes Played",
    y = "Points"
  ) +
  theme_minimal()
```



```r
# 3. Compute correlations
pearson  <- cor(nba_2017$MP, nba_2017$PTS, method = "pearson")
spearman <- cor(nba_2017$MP, nba_2017$PTS, method = "spearman")

pearson
```

```
## [1] 0.9038648
```

```r
spearman
```

```
## [1] 0.9686586
```

```r
cor.test(nba_2017$MP, nba_2017$PTS, method = "pearson")
```

```
##
##  Pearson's product-moment correlation
##
## data:  nba_2017$MP and nba_2017$PTS
## t = 51.144, df = 586, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.8878987 0.9176561
## sample estimates:
##       cor
## 0.9038648
```

```r
cor.test(nba_2017$MP, nba_2017$PTS, method = "spearman")
```

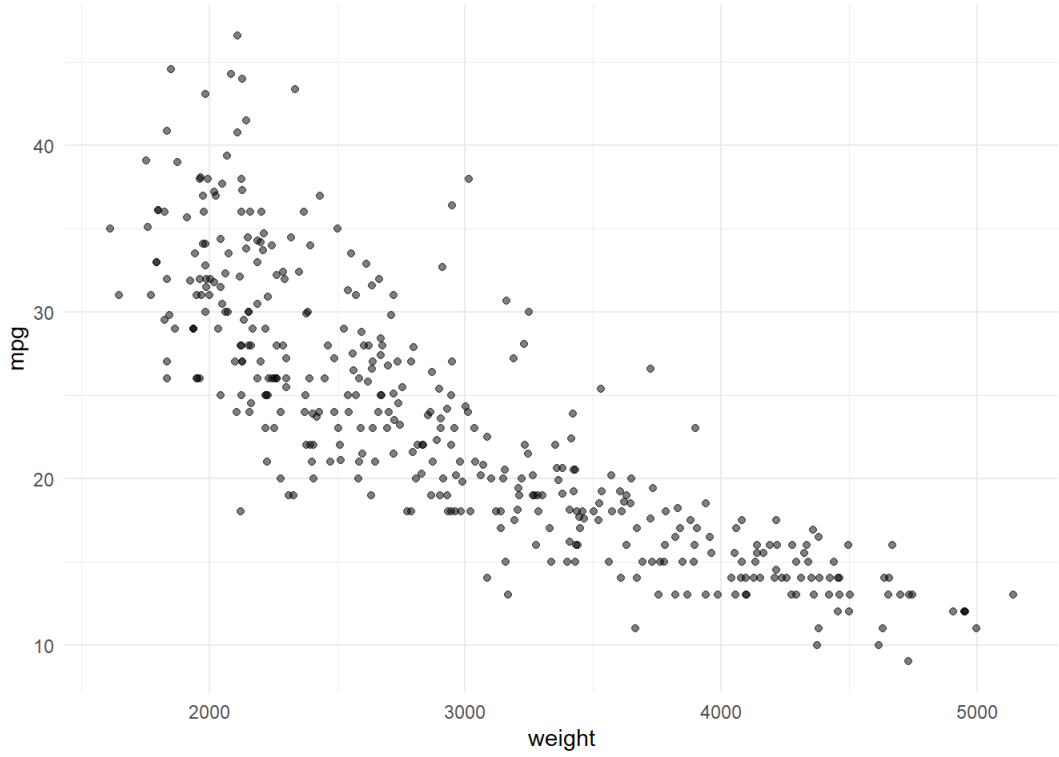
```
## Warning in cor.test.default(nba_2017$MP, nba_2017$PTS, method = "spearman"):
## Cannot compute exact p-value with ties
```

```
##
##  Spearman's rank correlation rho
##
## data:  nba_2017$MP and nba_2017$PTS
## S = 1061934, p-value < 2.2e-16
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.9686586
```

Ejercicio 2 - Cars

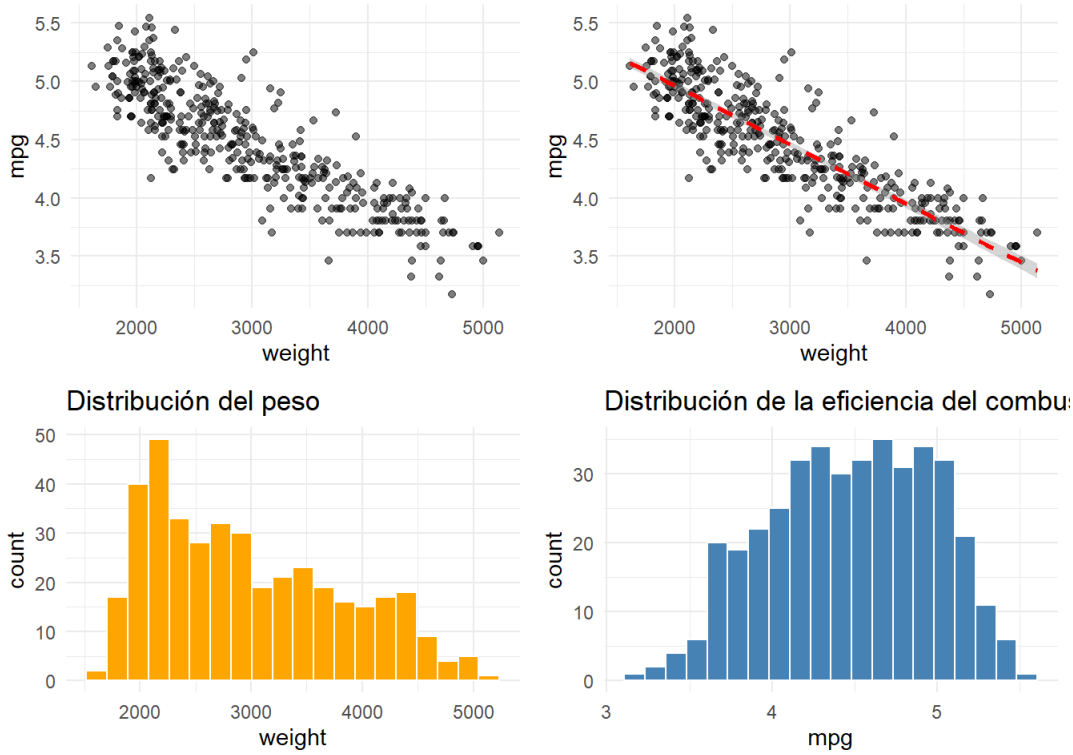
Car Weight vs Fuel Efficiency

```r
data <- read.csv("C:/Users/enriq/Desktop/Master/Data Science (MIB)/Clases Rafa/data/auto-mpg.csv", header
= T, sep = ",")

data1 <- data %>%
  select(mpg,weight)

ggplot(data1, aes(x = weight, y = mpg)) +
  geom_point(alpha = 0.5) +
  theme_minimal()
```




```
data1$mpg <- log2(data1$mpg)

Fig1 <- ggplot(data1, aes(x = weight, y = mpg)) +
  geom_point(alpha = 0.5) +
  theme_minimal()

Fig3 <- ggplot(data1, aes(weight)) +
  geom_histogram(bins = 20, fill = "orange", color = "white") +
  theme_minimal() +
  labs(title = "Distribución del peso")

Fig4 <- ggplot(data1, aes(mpg)) +
  geom_histogram(bins = 20, fill = "steelblue", color = "white") +
  theme_minimal() +
  labs(title = "Distribución de la eficiencia del combustible")

# Correleción

pearson <- cor(data1$weight, data1$mpg, method = "pearson")
spearman <- cor(data1$weight, data1$mpg, method = "spearman")

Fig2 <- ggplot(data1, aes(x = weight, y = mpg)) +
  geom_point(alpha = 0.5) +
  theme_minimal() +
  geom_smooth(method = "lm", color = "red", linetype = "dashed")

ggarrange(Fig1, Fig2, Fig3, Fig4)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



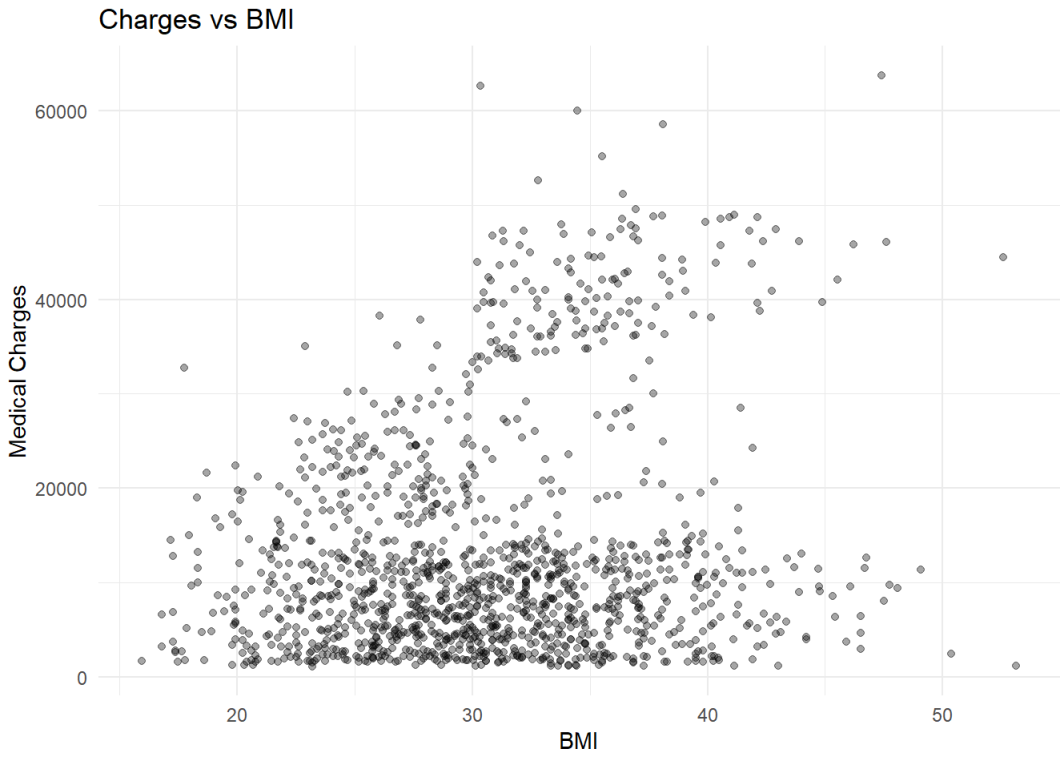

```
lm(mpg ~ weight, data = data1)
```

```
##
## Call:
## lm(formula = mpg ~ weight, data = data1)
##
## Coefficients:
## (Intercept)       weight
##   5.9792944   -0.0005066
```

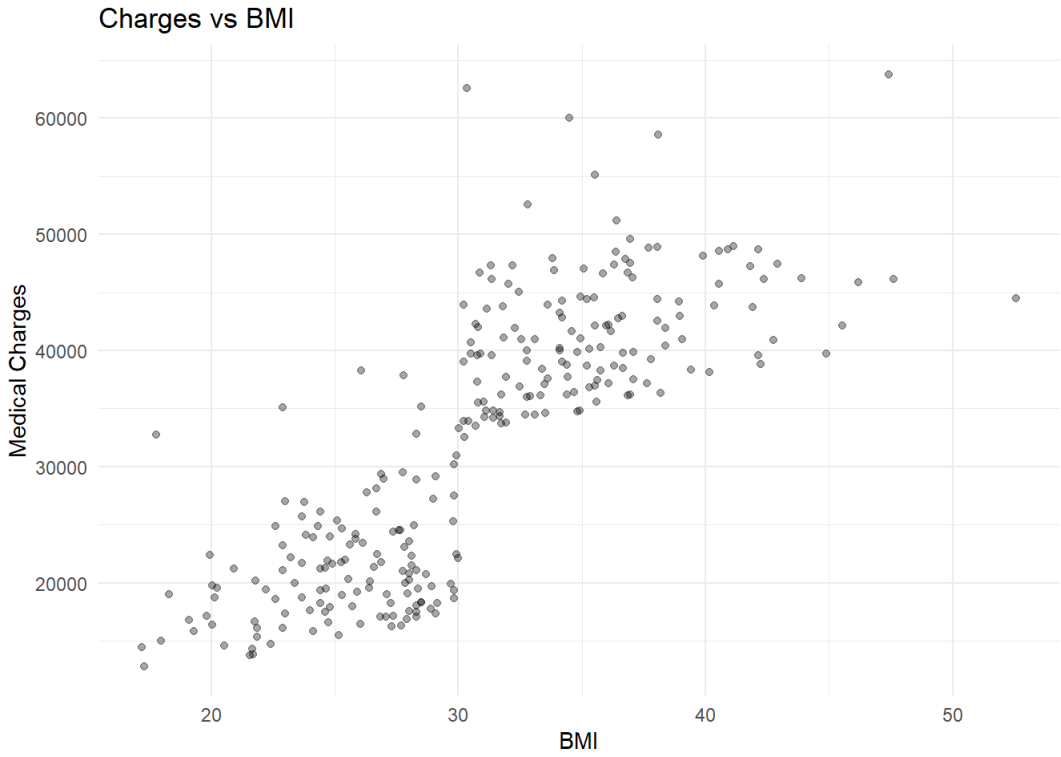
Ejercicio 3 - Insurance

BMI vs Medical charges

```
# Load and clean ------------------------------------------------------------
insurance <- read_csv("C:/Users/enriq/Desktop/Master/Data Science (MIB)/Clases Rafa/data/insurance.csv",
show_col_types = FALSE)

df <- subset(insurance)  %>%
  dplyr::select(charges, bmi) %>%
  dplyr::filter(is.finite(charges), is.finite(bmi))

ggplot(df, aes(bmi, charges)) +
  geom_point(alpha = 0.35) +
  labs(title = "Charges vs BMI", x = "BMI", y = "Medical Charges") +
  theme_minimal()
```

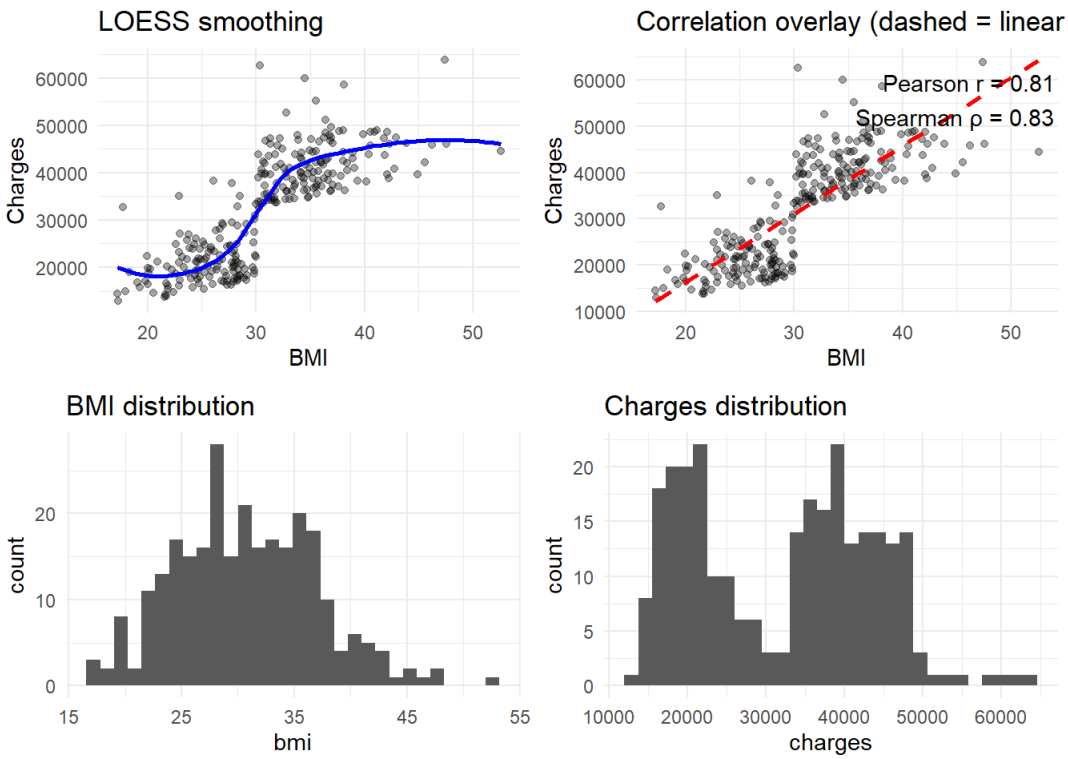



```
## any problem?
# check other colums!

# 1a) Scatter (shape of relationship)
df <- subset(insurance, smoker=="yes")  %>%
  dplyr::select(charges, bmi) %>%
  dplyr::filter(is.finite(charges), is.finite(bmi))

# 4) Visualize explained vs unexplained ------------------------------------

ggplot(df, aes(bmi, charges)) +
  geom_point(alpha = 0.35) +
  labs(title = "Charges vs BMI", x = "BMI", y = "Medical Charges") +
  theme_minimal()
```



```r
# 1b) Distributions
p_hist_bmi <- ggplot(df, aes(bmi)) +
  geom_histogram(bins = 30) +
  labs(title = "BMI distribution") +
  theme_minimal()

p_hist_charges <- ggplot(df, aes(charges)) +
  geom_histogram(bins = 30) +
  labs(title = "Charges distribution") +
  theme_minimal()

# 1c) Correlation
pearson  <- cor(df$bmi, df$charges, method = "pearson")
spearman <- cor(df$bmi, df$charges, method = "spearman")

p_scatter_loess <- ggplot(df, aes(bmi, charges)) +
  geom_point(alpha = 0.35) +
  geom_smooth(se = FALSE, method = "loess", color = "blue") +
  labs(title = "LOESS smoothing", x = "BMI", y = "Charges") +
  theme_minimal()

p_scatter_lm <- ggplot(df, aes(bmi, charges)) +
  geom_point(alpha = 0.35) +
  geom_smooth(se = FALSE, method = "lm", linetype = "dashed", color="red") +
  annotate("text", x = Inf, y = Inf, hjust = 1.02, vjust = 1.5,
           label = sprintf("Pearson r = %.2f\nSpearman ρ = %.2f", pearson, spearman)) +
  labs(title = "Correlation overlay (dashed = linear fit)", x = "BMI", y = "Charges") +
  theme_minimal()

ggpubr::ggarrange(p_scatter_loess, p_scatter_lm, p_hist_bmi, p_hist_charges,
                  ncol = 2, nrow = 2)
```

```
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
```

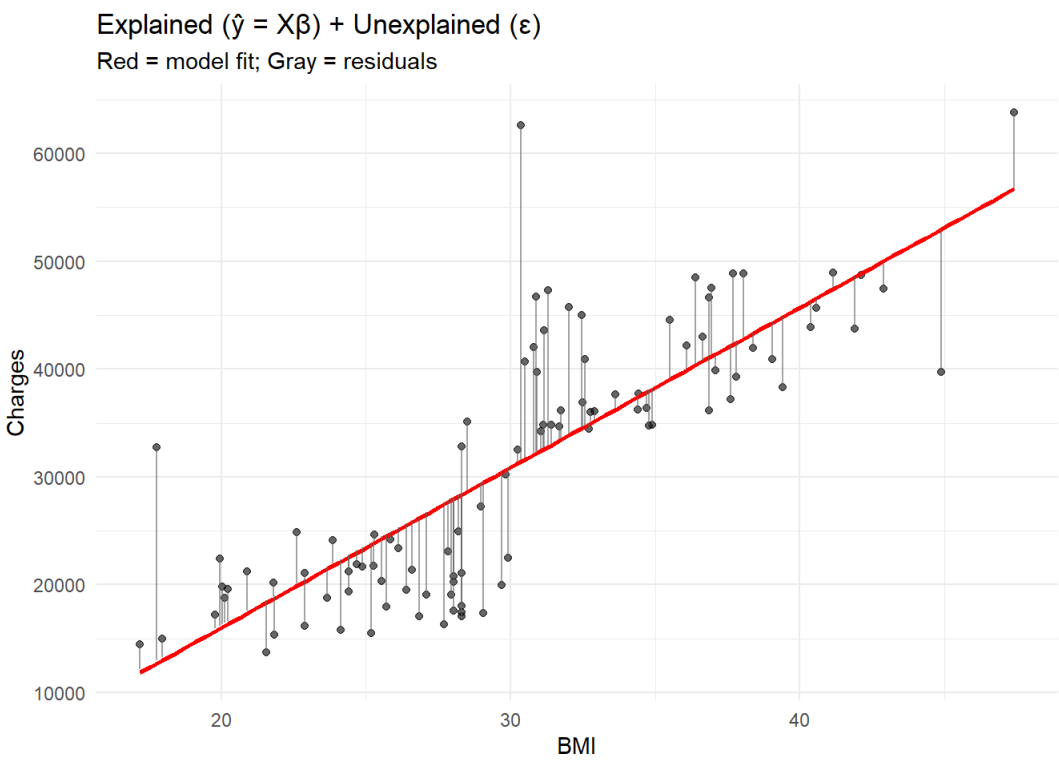



```r
# 2) Linear model ------------------------------------------------------------
m <- lm(charges ~ bmi, data = df)
summary(m)
```

```
##
## Call:
## lm(formula = charges ~ bmi, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -19768.0  -4487.9     34.4   3263.9  31055.9
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -13186.58    2052.88  -6.423 5.93e-10 ***
## bmi           1473.11      65.48  22.496  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6837 on 272 degrees of freedom
## Multiple R-squared:  0.6504, Adjusted R-squared:  0.6491
## F-statistic: 506.1 on 1 and 272 DF,  p-value: < 2.2e-16
```

```
df_mod <- augment(m)

set.seed(1)
samp <- df_mod %>% slice_sample(n = 100)

ggplot(samp, aes(bmi, charges)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  geom_segment(aes(xend = bmi, y = .fitted, yend = charges),
               color = "gray40", alpha = 0.6) +
  labs(title = "Explained (ŷ = Xβ) + Unexplained (ε)",
       subtitle = "Red = model fit; Gray = residuals",
       x = "BMI", y = "Charges") +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
# 3) Assumption checks -------------------------------------------------------

# Homoscedasticity
bptest(m)
```

```
##
##  studentized Breusch-Pagan test
##
## data:  m
## BP = 2.3496, df = 1, p-value = 0.1253
```

```
set.seed(1)
shapiro.test(sample(residuals(m), size = min(5000, length(residuals(m)))))
```

```
##
##  Shapiro-Wilk normality test
##
## data:  sample(residuals(m), size = min(5000, length(residuals(m))))
## W = 0.97665, p-value = 0.0001843
```

Ejercicio 4 - Insurance v2

Updated BMI vs Medical charges

```
# Load data
ins <- read_csv("C:/Users/enriq/Desktop/Master/Data Science (MIB)/Clases Rafa/data/insurance.csv",show_co
l_types = FALSE)

# Convert categorical variables
ins <- ins %>%
  mutate(
    smoker = factor(smoker),
    sex    = factor(sex)
  )

# 1) Single predictors
m1 <- lm(charges ~ bmi, data=ins)
m2 <- lm(charges ~ age, data=ins)
m3 <- lm(charges ~ smoker, data=ins)

# 2) Multiple predictors
m4 <- lm(charges ~ age + bmi + smoker, data=ins)

# Compare adjusted R²
glance(m1)$adj.r.squared
```

```
## [1] 0.03862008
```

```
glance(m2)$adj.r.squared
```

```
## [1] 0.08872432
```

```
glance(m3)$adj.r.squared
```

```
## [1] 0.6194802
```

```
glance(m4)$adj.r.squared
```

```
## [1] 0.7469093
```

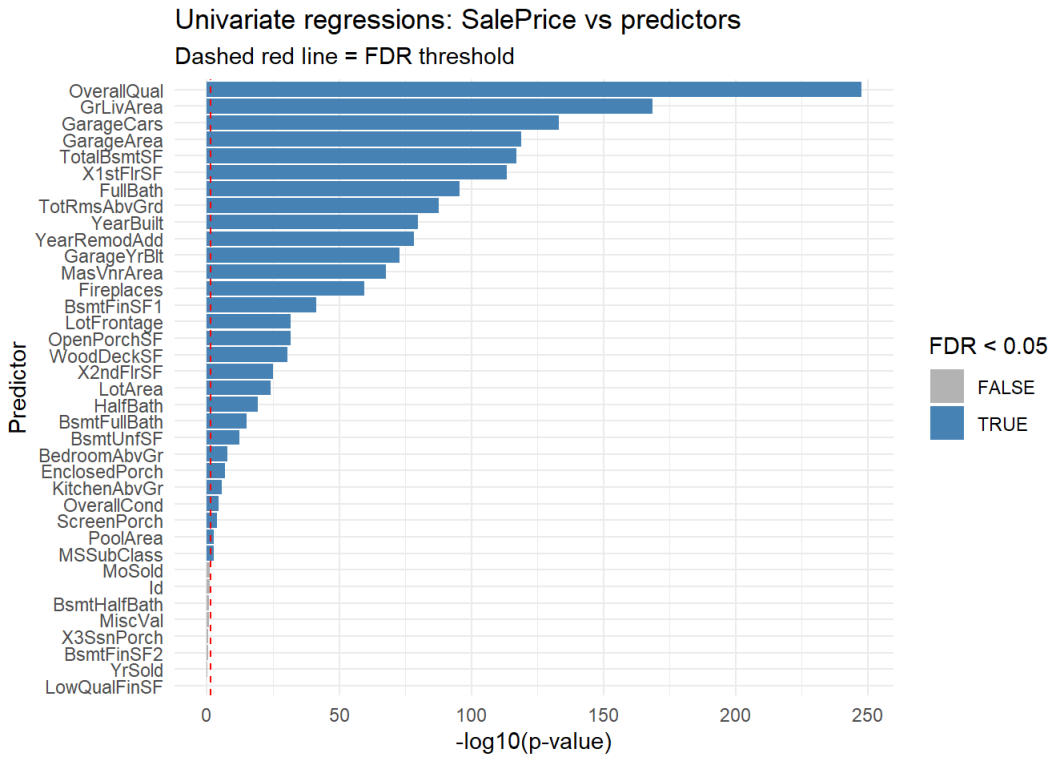
```
# Summaries
summary(m4)
```

```
##
## Call:
## lm(formula = charges ~ age + bmi + smoker, data = ins)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12415.4  -2970.9   -980.5   1480.0  28971.8
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -11676.83     937.57  -12.45   <2e-16 ***
## age            259.55      11.93   21.75   <2e-16 ***
## bmi            322.62      27.49   11.74   <2e-16 ***
## smokeryes    23823.68     412.87   57.70   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6092 on 1334 degrees of freedom
## Multiple R-squared:  0.7475, Adjusted R-squared:  0.7469
## F-statistic:  1316 on 3 and 1334 DF,  p-value: < 2.2e-16
```

Ejercicio 5 - FDR

FDR Correction

```r
# Load data (train.csv from Kaggle)
house <- read_csv("C:/Users/enriq/Desktop/Master/Data Science (MIB)/Clases Rafa/data/train.csv", show_col
_types = FALSE)

# Keep only numeric predictors
num_vars <- house %>%
  dplyr::select(where(is.numeric)) %>%
  drop_na()

names(num_vars) <- make.names(names(num_vars))

# Outcome
outcome <- "SalePrice"
predictors <- setdiff(names(num_vars), outcome)

# Univariate regressions
results_list <- list()

for (var in predictors) {
  # Build formula
  form <- as.formula(paste(outcome, "~", var))

  # Fit linear model
  m <- lm(form, data = num_vars)

  # Tidy output
  tmp <- tidy(m)

  # Drop intercept row
  tmp <- tmp[tmp$term != "(Intercept)", ]

  # Add predictor name
  tmp$predictor <- var

  # Store in list
  results_list[[var]] <- tmp
}

# Combine all results
results <- bind_rows(results_list)

# FDR correction
results <- results %>%
  dplyr::mutate(p_adj_fdr = p.adjust(p.value, method="fdr"),
                neglog10p = -log10(p.value),
                significant = p_adj_fdr < 0.05) %>%
  dplyr::arrange(p_adj_fdr)

# plot
ggplot(results, aes(x = reorder(predictor, neglog10p), y = neglog10p, fill = significant)) +
  geom_col() +
  coord_flip() +
  geom_hline(yintercept = -log10(0.05), color = "red", linetype = "dashed") +
  scale_fill_manual(values=c("gray70", "steelblue")) +
  labs(
    title = "Univariate regressions: SalePrice vs predictors",
    subtitle = "Dashed red line = FDR threshold",
    x = "Predictor",
    y = "-log10(p-value)",
    fill = "FDR < 0.05"
  ) +
  theme_minimal()
```

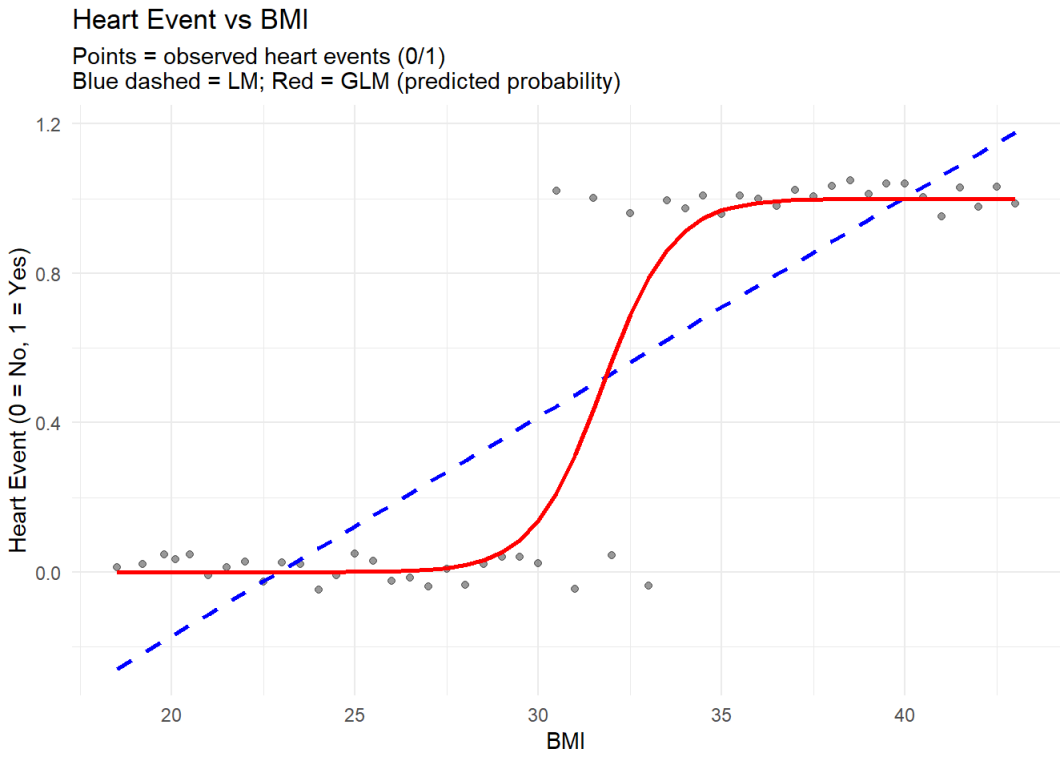


Ejercicio 6 - BMI Heart Attack

GLMs

```r
# Artificial dataset: BMI vs Heart Event (0/1)
df <- data.frame(
  BMI = c(
    18.5, 19.2, 19.8, 20.1, 20.5,
    21.0, 21.5, 22.0, 22.5, 23.0,
    23.5, 24.0, 24.5, 25.0, 25.5,
    26.0, 26.5, 27.0, 27.5, 28.0,
    28.5, 29.0, 29.5, 30.0, 30.5,
    31.0, 31.5, 32.0, 32.5, 33.0,
    33.5, 34.0, 34.5, 35.0, 35.5,
    36.0, 36.5, 37.0, 37.5, 38.0,
    38.5, 39.0, 39.5, 40.0, 40.5,
    41.0, 41.5, 42.0, 42.5, 43.0
  ),
  Event = c(
    0,0,0,0,0,
    0,0,0,0,0,
    0,0,0,0,0,
    0,0,0,0,0,
    0,0,0,0,1,
    0,1,0,1,0,
    1,1,1,1,1,
    1,1,1,1,1,
    1,1,1,1,1,
    1,1,1,1,1
  )
)

# Fit linear regression (bad for binary) and logistic regression (correct)
lm_fit  <- lm(Event ~ BMI, data = df)
glm_fit <- glm(Event ~ BMI, data = df, family = binomial(link = "logit"))
# family binomial - binary events / logistic as the link function

df$lm_pred  <- predict(lm_fit, type = "response")
df$glm_pred <- predict(glm_fit, type = "response")

# Plot
ggplot(df, aes(BMI, Event)) +
  geom_jitter(height = 0.05, width = 0, alpha = 0.4) +
  geom_line(aes(y = lm_pred), color = "blue", size = 1, linetype = "dashed") +
  geom_line(aes(y = glm_pred), color = "red", size = 1) +
  labs(title = "Heart Event vs BMI",
       subtitle = "Points = observed heart events (0/1)\nBlue dashed = LM; Red = GLM (predicted probabili
ty)",
       y = "Heart Event (0 = No, 1 = Yes)",
       x = "BMI") +
  theme_minimal()
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## ℹ Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```




```
# probabilities vs actual events
sum(abs(df$lm_pred  - df$Event))
```

```
## [1] 10.75468
```

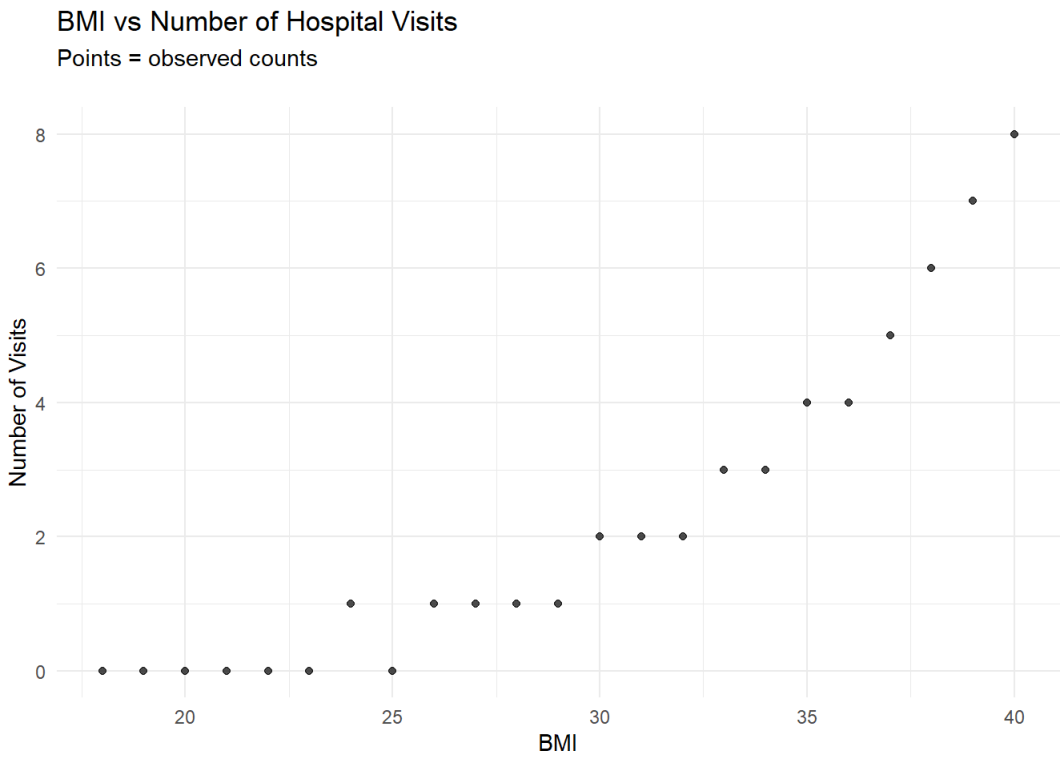
```
sum(abs(df$glm_pred - df$Event))
```

```
## [1] 4.0411
```

Ejercicio 7 - Visitas Hospital

Visits vs BMI / Poisson

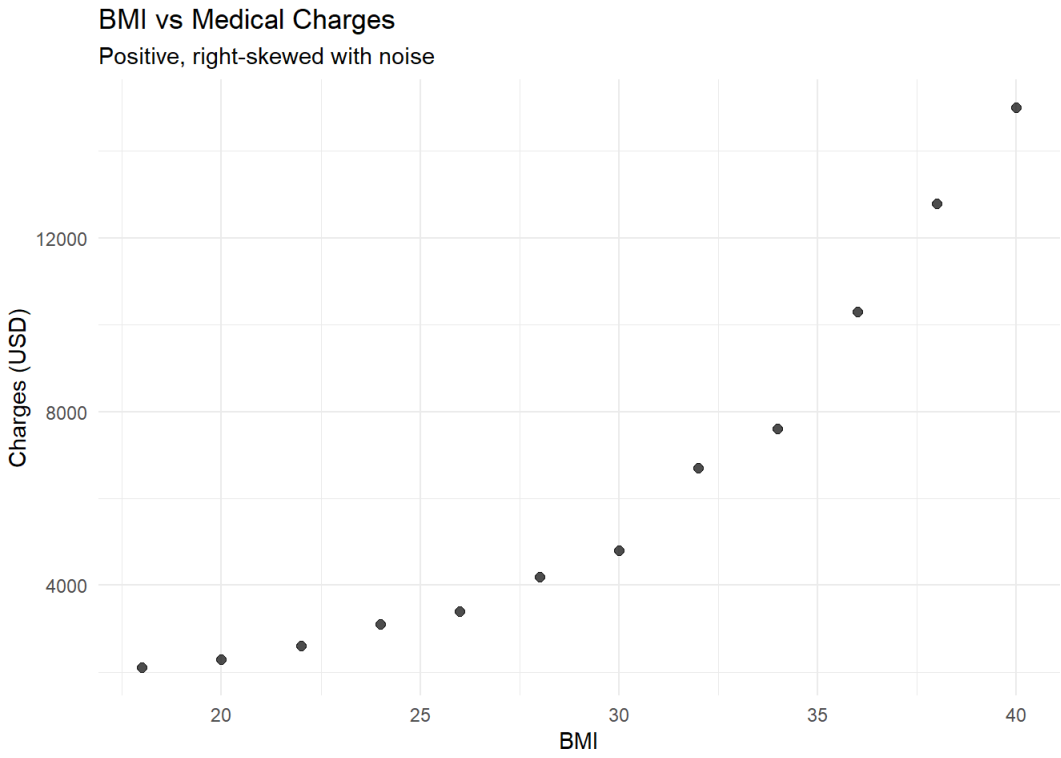
```
df <- data.frame(
  BMI = c(
    18, 19, 20, 21, 22,
    23, 24, 25, 26, 27,
    28, 29, 30, 31, 32,
    33, 34, 35, 36, 37,
    38, 39, 40
  ),
  Visits = c(
    0, 0, 0, 0, 0,    # 18-22
    0, 1, 0, 1, 1,    # 23-27
    1, 1, 2, 2, 2,    # 28-32
    3, 3, 4, 4, 5,    # 33-37
    6, 7, 8           # 38-40
  )
)

ggplot(df, aes(BMI, Visits)) +
  geom_point(alpha = 0.7) +
  labs(title = "BMI vs Number of Hospital Visits",
       subtitle = "Points = observed counts\n",
       y = "Number of Visits",
       x = "BMI") +
  theme_minimal()
```



Ejercicio 8 - BMI Charges

BMI vs Charges

```
# BMI vs Charges
df <- data.frame(
  BMI = c(18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40),
  Charges = c(2100, 2300, 2600, 3100, 3400,
              4200, 4800, 6700, 7600, 10300,
              12800, 15000)
)

# Scatter plot
ggplot(df, aes(BMI, Charges)) +
  geom_point(size = 2, alpha = 0.7) +
  labs(title = "BMI vs Medical Charges",
       subtitle = "Positive, right-skewed with noise",
       x = "BMI", y = "Charges (USD)") +
  theme_minimal()
```


BMI vs Medical Charges
Positive, right-skewed with noise
Charges (USD)
12000
8000
4000
20
25
30
35
40
BMI